

# Logistic Regression

egg 2026, experimental semantics, day 5

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## 1 Logistic regression

- based on [The Art of Statistics GitHub repo](#)

### 1.1 Classical linear regression

Data from 1991-1995 are contained in [05-1-galton-x.csv](#). Although the book says the data is from [HistData: Data Sets from the History of Statistics and Data Visualization, 2018](#), this lesson uses this version of [Galton's Height Data](#).

```
galton <- read.csv("data/05-1-galton-x.csv", header = TRUE)
attach(galton)

summary(galton)
```

| Family        | Father        | Mother        | Gender       |
|---------------|---------------|---------------|--------------|
| Length :898   | Min. :62.00   | Min. :58.00   | Length :898  |
| N.unique :197 | 1st Qu.:68.00 | 1st Qu.:63.00 | N.unique : 2 |
| N.blank : 0   | Median :69.00 | Median :64.00 | N.blank : 0  |
| Min.nchar: 1  | Mean :69.23   | Mean :64.08   | Min.nchar: 1 |
| Max.nchar: 4  | 3rd Qu.:71.00 | 3rd Qu.:65.50 | Max.nchar: 1 |
|               | Max. :78.50   | Max. :70.50   |              |

| Height        | Kids           |
|---------------|----------------|
| Min. :56.00   | Min. : 1.000   |
| 1st Qu.:64.00 | 1st Qu.: 4.000 |
| Median :66.50 | Median : 6.000 |
| Mean :66.76   | Mean : 6.136   |
| 3rd Qu.:69.70 | 3rd Qu.: 8.000 |
| Max. :79.00   | Max. :15.000   |

```

Unique.Fathers <- numeric()
Unique.Mothers <- numeric()
nunique <- 1
Unique.Fathers[1] <- Father[1]
Unique.Mothers[1] <- Mother[1]

for (i in 2:length(Family)) {
  if (Family[i] != Family[i - 1]) {
    nunique <- nunique + 1
    Unique.Fathers[nunique] <- Father[i]
    Unique.Mothers[nunique] <- Mother[i]
  }
}

length(Unique.Fathers)

```

```
[1] 197
```

```
summary(Unique.Fathers)
```

| Min.  | 1st Qu. | Median | Mean  | 3rd Qu. | Max.  |
|-------|---------|--------|-------|---------|-------|
| 62.00 | 68.00   | 69.50  | 69.35 | 71.00   | 78.50 |

```
sd(Unique.Fathers)
```

```
[1] 2.622034
```

```
length(Unique.Mothers)
```

```
[1] 197
```

```
summary(Unique.Mothers)
```

| Min.  | 1st Qu. | Median | Mean  | 3rd Qu. | Max.  |
|-------|---------|--------|-------|---------|-------|
| 58.00 | 62.70   | 64.00  | 63.98 | 65.50   | 70.50 |

```
sd(Unique.Mothers)
```

```
[1] 2.355607
```

```
Son <- Height[Gender == "M"]  
length(Son)
```

```
[1] 465
```

```
summary(Son)
```

| Min.  | 1st Qu. | Median | Mean  | 3rd Qu. | Max.  |
|-------|---------|--------|-------|---------|-------|
| 60.00 | 67.50   | 69.20  | 69.23 | 71.00   | 79.00 |

```
sd(Son)
```

```
[1] 2.631594
```

```
Daughter <- Height[Gender == "F"]  
length(Daughter)
```

```
[1] 433
```

```
summary(Daughter)
```

| Min.  | 1st Qu. | Median | Mean  | 3rd Qu. | Max.  |
|-------|---------|--------|-------|---------|-------|
| 56.00 | 62.50   | 64.00  | 64.11 | 65.50   | 70.50 |

```
sd(Daughter)
```

```
[1] 2.37032
```

### 1.1.1 Figure 5.1 (page 124) Linear regression of sons' on fathers' heights

```
FatherS <- Father[Gender == "M"]
```

```
fit <- lm(Son ~ FatherS)
Predicted <- predict(fit)
summary(fit)
```

Call:

```
lm(formula = Son ~ FatherS)
```

Residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -9.3774 | -1.4968 | 0.0181 | 1.6375 | 9.3987 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | 38.25891 | 3.38663    | 11.30   | <2e-16 *** |
| FatherS     | 0.44775  | 0.04894    | 9.15    | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.424 on 463 degrees of freedom

Multiple R-squared: 0.1531, Adjusted R-squared: 0.1513

F-statistic: 83.72 on 1 and 463 DF, p-value: < 2.2e-16

```
FatherS.j <- jitter(FatherS, factor = 5)
```

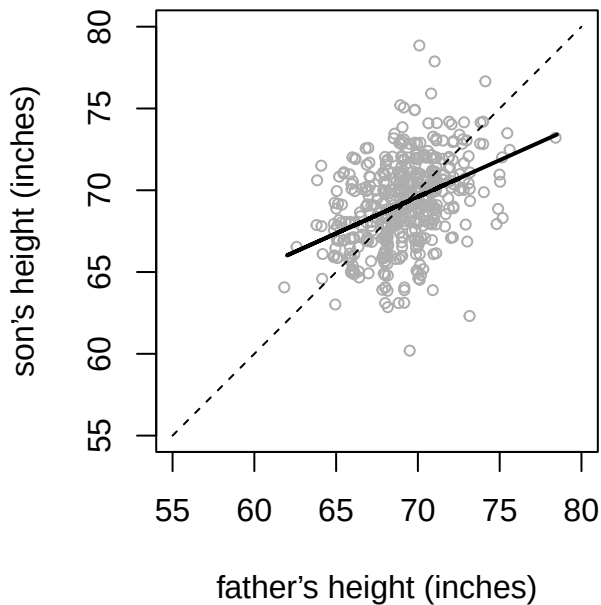
```
Son.j <- jitter(Son, factor = 5)
```

```
xlims <- ylims <- c(55, 80)
```

```
par(mfrow = c(1, 1), mar = c(4, 4, 2, 0), pty = "s")
```

```
plot(FatherS.j, Son.j, xlim = xlims, ylim = ylims, cex = 0.7,
     xlab = "father's height (inches)", ylab = "son's height (inches)", col = "gray68")
```

```
lines(c(xlims[1], xlims[2]), c(xlims[1], xlims[2]), lty = 2)
lines(Predicted ~ FatherS, lwd = 2)
```



### 1.1.2 Now in ggplot

```
library(ggplot2)

Males <- cbind.data.frame(FatherS, FatherS.j, Son, Son.j, Predicted)

p <- ggplot(Males, aes(x = FatherS, y = Son))
p <- p + geom_point(x = FatherS.j, y = Son.j, shape = 1)
p <- p + labs(x = "Father's height (inches)", y = "Son's height (inches)")
p <- p + theme(legend.position = "none")
p <- p + expand_limits(x = c(55, 80), y = c(55, 80))
p <- p + geom_line(aes(FatherS, Predicted), linewidth = 1.5)

p <- p + geom_abline(slope = 1, linetype = "dashed")

pointA <- c(137)
pointB <- c(28)

p <- p + geom_point(aes(x = FatherS.j[pointA], y = Predicted[pointA]), shape = 1)
```

```

p <- p + geom_point(aes(x = FatherS.j[pointA], y = Son.j[pointA]), shape = 1)
p <- p + geom_segment(linetype = "dashed", linewidth = 1, colour = "purple",
  aes(x = FatherS.j[pointA], y = Son.j[pointA],
    xend = FatherS.j[pointA], yend = Predicted[pointA]))

p <- p + geom_point(aes(x = FatherS.j[pointB], y = Predicted[pointB]), shape = 1)
p <- p + geom_point(aes(x = FatherS.j[pointB], y = Son.j[pointB]), shape = 1)
p <- p + geom_segment(linetype = "dashed", linewidth = 1, colour = "purple",
  aes(x = FatherS.j[pointB], y = Son.j[pointB],
    xend = FatherS.j[pointB], yend = Predicted[pointB]))

p

```

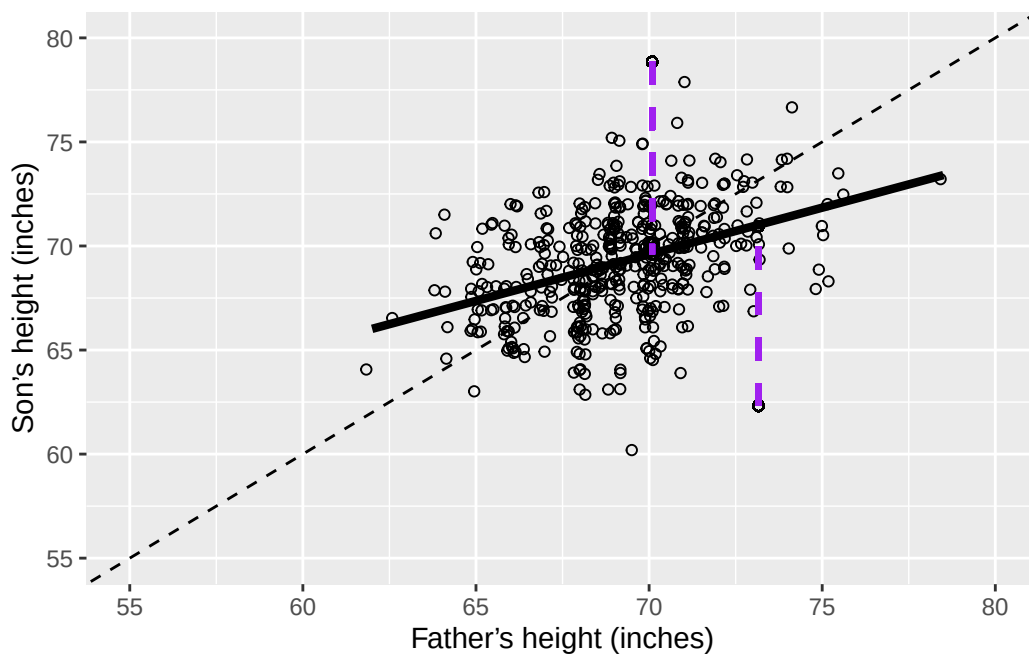


Figure 5.1 Scatter of heights of 465 fathers and sons from Galton's data (many fathers are repeated since they have multiple sons). A jitter has been added to separate the points, and the diagonal dashed line represents exact equality between son and father heights. The solid line is the standard best-fit line. Each point gives rise to a residual (dashed line), which is the size of the error if we use the line to predict a son's height from his father's.

## 1.2 Logistic regression

Data from 1991-1995 are contained in [02-5-child-heart-surgery-1991-x.csv](#), taken from D. J. Spiegelhalter et al., [Commissioned Analysis of Surgical Performance Using Routine Data:](#)

Lessons from the Bristol Inquiry.

```
library(ggplot2)

child.1991 <- read.csv("data/02-5-child-heart-surgery-1991-x.csv")

attach(child.1991)
fit <- glm(Survivors / Operations ~ Operations,
           weight = Operations,
           family = "binomial",
           data = child.1991[-1, ])
summary(fit)
```

Call:

```
glm(formula = Survivors/Operations ~ Operations, family = "binomial",
     data = child.1991[-1, ], weights = Operations)
```

Coefficients:

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.7348545   0.1410843  12.297   <2e-16 ***
Operations    0.0009615   0.0003807   2.526   0.0115 *
```

---

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 18.623  on 10  degrees of freedom
Residual deviance: 12.169  on  9  degrees of freedom
AIC: 72.622
```

Number of Fisher Scoring iterations: 4

```
predictions <- 100 * predict(fit, data.frame(Operations = 0:700), type = "response")
pred.frame <- data.frame(Extremes = 0:700, predictions)
```

```
p <- ggplot()
p <- p + geom_point(child.1991, mapping = aes(x = Operations, y = 100 * Survivors / Operations))
p <- p + expand_limits(x = c(0, 700), y = c(70, 100))
p <- p + labs(x = "Number of operations", y = "% 30-day survival", title = "(a) Survival in v")
p <- p + geom_line(data = pred.frame, aes(x = Extremes, y = predictions), linewidth = 1)
```

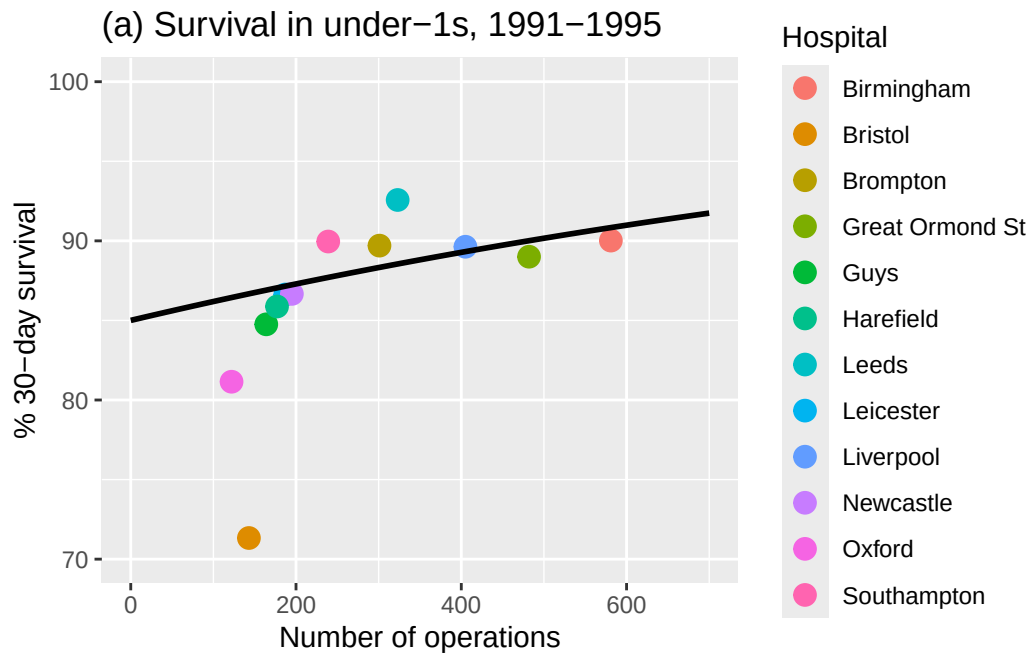


Figure 5.2 Fitted logistic regression model for child heart surgery data for under-1s in UK hospitals between 1991 and 1995. Hospitals treating more patients have better survival. The line is part of a curve that will never reach 100%, and is fitted ignoring the outlying data point representing Bristol.

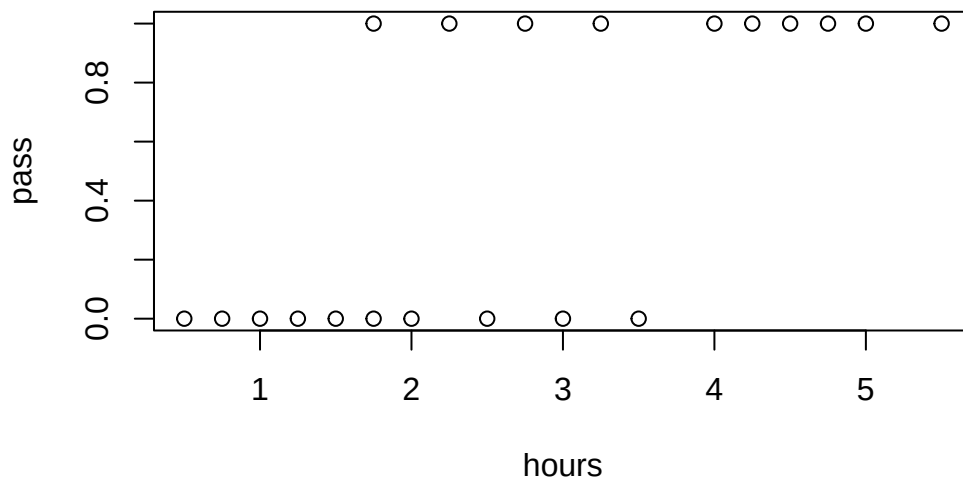
### 1.3 Another non-linguistic example

- from [Wikipedia](#)

```
hours <- c(0.50, 0.75, 1.00, 1.25, 1.50, 1.75, 1.75, 2.00, 2.25, 2.50,
          2.75, 3.00, 3.25, 3.50, 4.00, 4.25, 4.50, 4.75, 5.00, 5.50)

pass <- c(0, 0, 0, 0, 0, 0, 1, 0, 1, 0,
          1, 0, 1, 0, 1, 1, 1, 1, 1, 1)

plot(hours, pass)
```



```
df <- data.frame(hours, pass)

logitm <- glm(pass ~ hours, data = df, family = "binomial")

logitm
```

Call: glm(formula = pass ~ hours, family = "binomial", data = df)

Coefficients:

|             |       |
|-------------|-------|
| (Intercept) | hours |
| -4.078      | 1.505 |

Degrees of Freedom: 19 Total (i.e. Null); 18 Residual

Null Deviance: 27.73

Residual Deviance: 16.06 AIC: 20.06

```
summary(logitm)
```

Call:

glm(formula = pass ~ hours, family = "binomial", data = df)

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z ) |
|-------------|----------|------------|---------|----------|
| (Intercept) | -4.0777  | 1.7610     | -2.316  | 0.0206 * |
| hours       | 1.5046   | 0.6287     | 2.393   | 0.0167 * |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 27.726 on 19 degrees of freedom  
Residual deviance: 16.060 on 18 degrees of freedom  
AIC: 20.06

Number of Fisher Scoring iterations: 5

```
exp(1.5046)
```

```
[1] 4.502352
```

## 1.4 Linguistic example

- [Acquisition experiment](#)
- [Proceedings article](#)